COM Kitchens : An Unedited Overhead-view Video Dataset as a Vision-Language Benchmark

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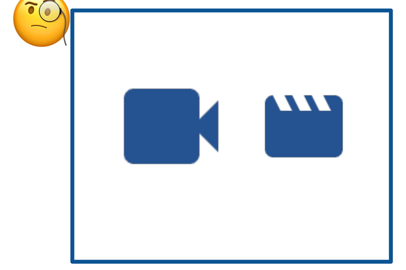
Created **Procedural Understanding** Research Platform for **Unedited User-captured Videos!**

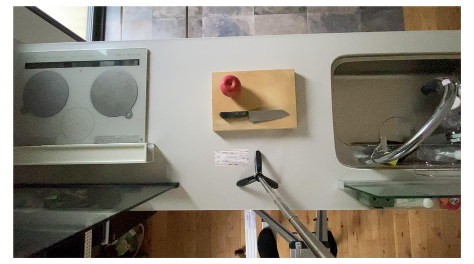
Motivation

We focus on real-world applications.

Procedural video understanding relies on curated web videos. However, users capture **unedited, fixed-viewpoint videos with their smartphones.**



Web-crawled
Ego-centric
Edited



✔ User-captured
✔ Fixed-View
✔ Unedited

COM Kitchens Dataset

Video Collection Procedure

Recipe Selection

Selected recipes that take <30 min from Cookpad Recipe Dataset.

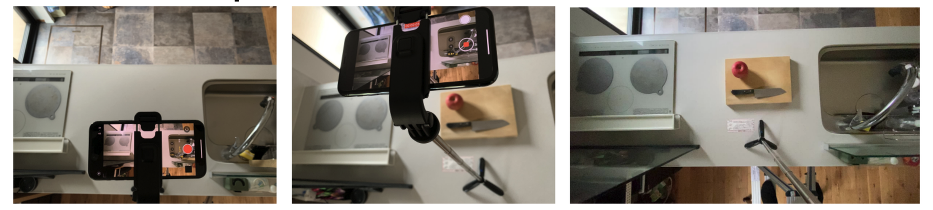
Filming

Recorded with iPhone 11 Pro fixed to a tripod.

Quality Control

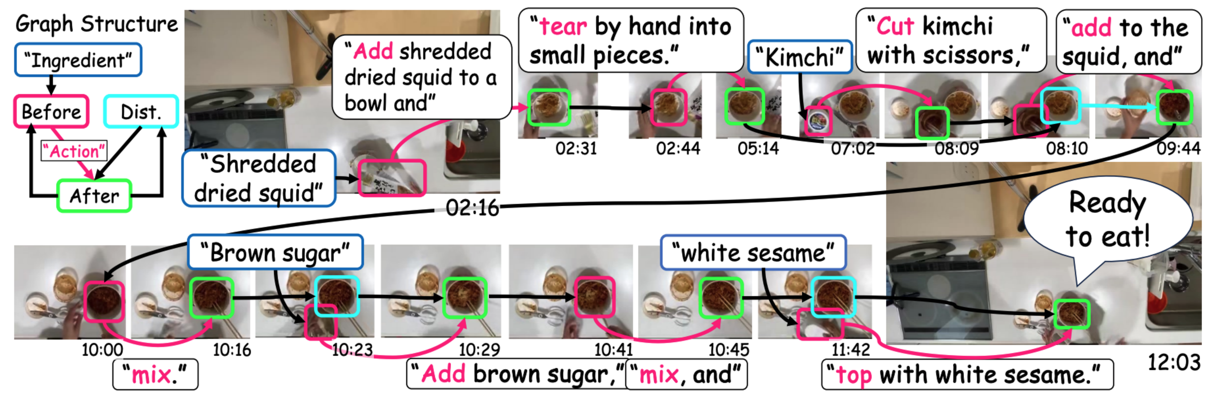
Exclude 200/410 videos that ignored instructions

Example of the film set and recorded content



Visual Action Graph (VAG)

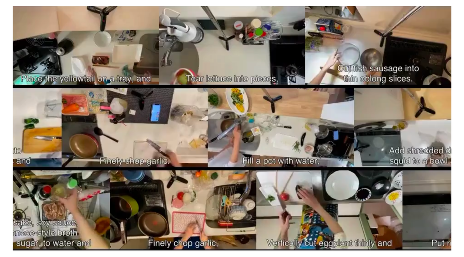
VAG describes entire flow of cooking task.



Specialists **annotated VAGs** from Feb. to Sep. 2023 (430 hrs).

Dataset Characteristics

70 Env.
139 Recipes
145 Movies
40 hours
6,826 bboxes
8,061 edges



Comparison between **instructional** video datasets with **FV** cameras.

dataset	year	topic	tasks	# env.	# videos	total (h)	avg. (m)	seg. type	seg. description
MMAC [45]	2008	Cooking	1	1	32	8	15.0	action	130 actions cls.
MPII [39]	2012	Cooking	14	1	44	8	13.4	action	65 actions cls.
ACE [43]	2012	Cooking	5	1	35	2	3.6	action	8 actions cls.
50 salads [46]	2013	Cooking	2	1	50	5	5.4	action	51 actions cls.
Breakfast [22]	2014	Cooking	-	18	1,712	77	2.7	action	10 actions cls.
IKEA ASM [3]	2021	Furniture	4	5	371	35	5.7	action	noun+verb (n+v)
Assembly101 [40]	2022	Assembly	15	1	4,321	513	7.1	act./step	1,380 act. cls./n+v
COM Kitchens	Ours	Cooking	139	70	145	40	16.6	act./step	instructional text

✔ Various env. and tasks
✔ Long durations

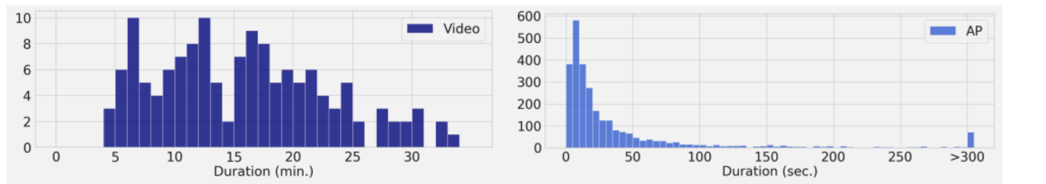
✔ The only fixed-view V&L video dataset

Comparison between **instructional vision-language video** datasets.

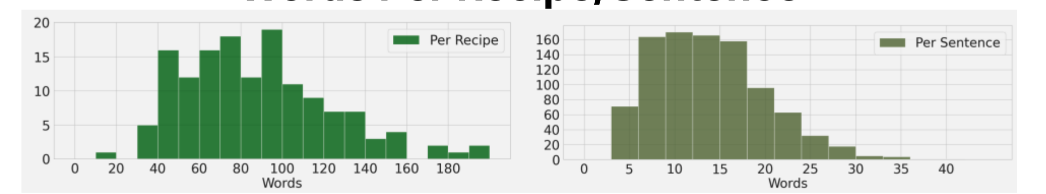
dataset	year	type	topic	tasks	# videos	total (h)	avg. (m)	seg. description	interval	# seg.
YouCookII [57]	2018	Web	Cook.	89	2,000	176	5.3	coarse instruction	manual	4,325
ProceL [10]	2019	Web	Multi.	12	720	47	3.9	coarse instruction	manual	498
COIN [47]	2019	Web	Multi.	180	11,827	476	2.4	coarse instruction	manual	46,354
CrossTask [58]	2019	Web	Multi.	83	4,700	376	4.8	coarse instruction	manual	19,278
MMAC-Captions [33]	2021	Ego	Cook.	5	146	16	13.4	coarse instruction	manual	5,002
Epic Kitchens [7]	2022	Ego	Cook.	70	700	100	8.6	narration	utterance	39,596
Ego4D [14]	2022	Ego	Open	-	-	3,670	-	narration	manual*	-
BioVL2 [38]	2022	Ego	Bio.	5	32	3	5.3	fine instruction	manual	408
VRF [44]	2022	Web	Cook.	200	200	2	0.7	fine instruction	manual	3,705
FineBio [51]	2024	Ego	Bio.	7	226	14.5	3.9	fine instruction	manual	3,541
COM Kitchens	Ours	FV	Cook.	139	145	40	16.6	fine instruction	manual	2,852

Dataset Statistics

Durations per Video/Action-by-person(AP)



Words Per Recipe/Sentence



APs **Ingredients**

Reasons For Refusal

Reason	# of Refusals	% among Refusals
1. Inappropriate view (e.g., stove is not covered)	97	50.5%
2. Faces in the view	46	24.0%
3. Skipped steps using pre-processed food	12	6.3%
4. Overly complicated process	11	5.7%
5. Pause and resume in recording	9	3.1%
6. Recording by slow mode	5	2.6%
7. Personal documents in the view	3	1.6%
8. Any other reasons	10	3.1%

Task I : Dense Video Captioning on Unedited Overhead-view Videos

Purpose

Task Objectives

DVC-OV is an offline cross-modal task that generates instructional text from demonstrations.

Challenges

Domain gap between unedited fixed-view videos and traditional DVC targets such as web videos.

Future Applications

Accessibility, content analysis, and enhanced video search capabilities.

Settings

Input: Entire overhead-view video.
Output: Captions to a specific AP.
Metrics: SODA_c, CIDEr, METEOR
Models: PDVC, Vid2Seq
Ablation: w or w/o fine-tuning, RL, AS

Experiments & Results

Can SoTA models solve the DVC-OV task?
Can models leverage action graph as **relation labels** (RL) or **attention supervision** (AS), used in current DVC approach?

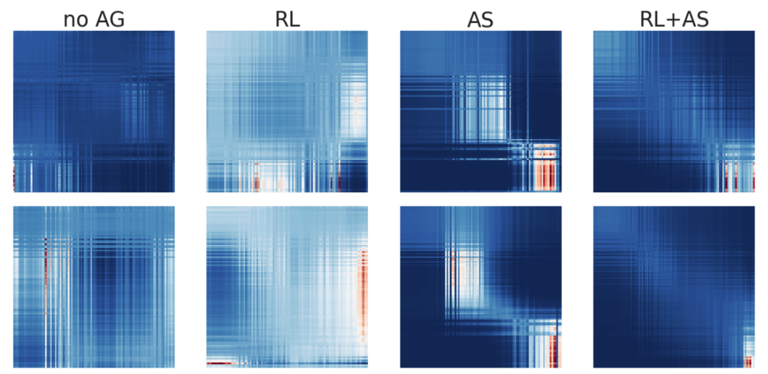
Zero-shot performance of existing DVC models is low, indicating a gap between web and user-captured videos.
Action graph during fine-tuning improves caption quality.
Attention Supervision (**AS**) and Relation Labels (**RL**) works!

COM Kitchens dataset is challenging for SoTAs.

DVC performances of baseline models

Model	FT	AG	SODA_c(↑)	CIDEr(↑)	METEOR(↑)
PDVC [49]	-	-	0.022	0.000	0.000
Vid2Seq [53]	-	-	0.017	0.066	0.010
Vid2Seq	✓	-	0.369	2.832	0.642
Vid2Seq	✓	RL	0.211	1.381	0.285
Vid2Seq	✓	AS	0.266	2.513	0.423
Vid2Seq	✓	RL+AS	0.581	6.195	1.142

Attention in the last encoder layer



Task II : Online Recipe Retrieval

Purpose

Task Objectives

Online Recipe Retrieval is an online cross-modal task for retrieving recipes **during** cooking.

Challenges

Compared to video-to-text retrieval, **OnRR** requires both **classification of video type** and **comprehension of the cooking process.**

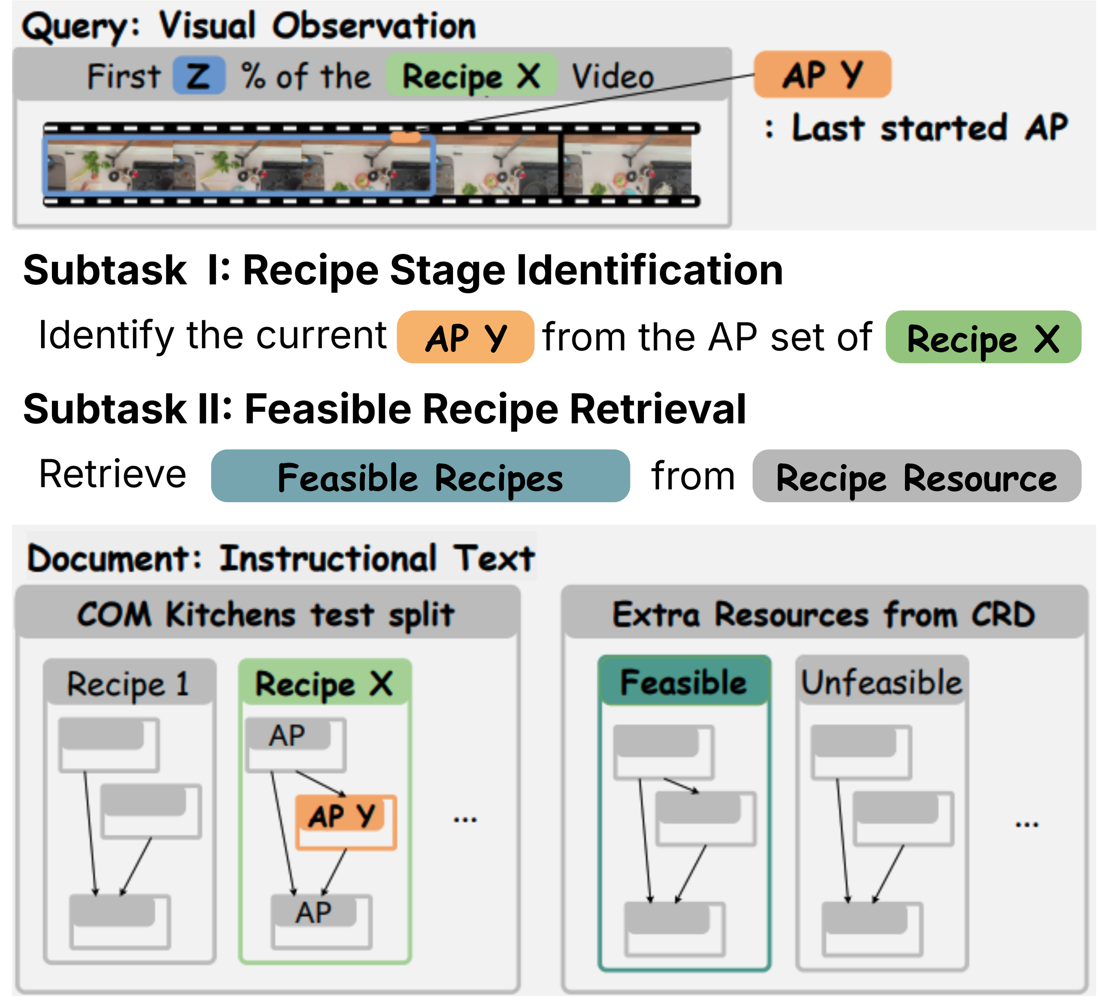
Future Applications

Develop practical smartphone applications for **online recipe recommendation.**

Settings

Metrics: Recall(**R@K**), Median Rank (**MdR**)
Models: UniVL, CLIP4Clip, X-CLIP, Random
Ablation: Fine-tuning on COM Kitchens

Task Overview



Results

Can SoTA models solve the OnRR task?

None of the model beats random selection in feasible recipe retrieval.
Fine-tuning improved the recipe stage retrieval performances.

Contrastive learning cannot solve two tasks simultaneously.

OnRR performances of baseline models

Task	Method	Early (25%) R@1	Early (25%) R@5	Early (25%) R@10	Early (25%) MdR	Middle (50%) R@1	Middle (50%) R@5	Middle (50%) R@10	Middle (50%) MdR
Feasible Recipe Retrieval	Random	1.8	8.6	15.8	-	0.4	1.8	3.1	-
	UniVL [25]	3.4	5.7	9.2	227.0	3.4	5.7	9.2	231.0
	CLIP4Clip [26]	0.0	0.0	10.3	79.0	0.0	0.0	6.8	85.0
	X-CLIP [27]	0.0	6.8	10.3	89.0	0.0	3.4	3.4	320.0
Recipe Stage Identification	Random	6.3	31.6	63.3	8.0	6.3	31.6	63.3	8.0
	UniVL [25]	17.2	48.2	68.9	5.0	9.2	63.3	89.2	3.0
	CLIP4Clip [26]	6.8	48.2	68.9	5.0	10.3	55.1	86.2	4.0
	X-CLIP [27]	10.3	51.7	68.9	4.0	17.2	37.9	93.1	6.0

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